An Exploratory Analysis of Machine Learning for Stock Market Predictions: Using the Power Set of Relevant Features to Enhance Performance

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Objective

Background: Predicting future prices of stocks in large markets is an ambiguous problem in which Machine Learning models have been used extensively

Competition Goal: Rank stocks by expected return and choose a portfolio that maximizes return and determine which financial features best predict prices

Competition evaluation: Sharpe Ratio of the difference in returns if we bought the top 200 stock and shorted the bottom 200 stock starting the next day and sold the day after

Data: We leverage a Kaggle Dataset with Japan Stock Exchange Data

kaggle **JPX Tokyo Stock Exchange Prediction**

$$Sharpe\ Ratio = \frac{Average(R_{day_1 - day_x})}{STD(R_{day_1 - day_x})}$$

Our approach: Train cutting-edge models using every non-empty subset from the power set of relevant features, extracted using PCA from a full feature set

Hypothesis: (1) Trying all subsets after PCA will provide a "best-performing" subset for each model better than using the full set of features without PCA, and (2) the best subsets will differ among models

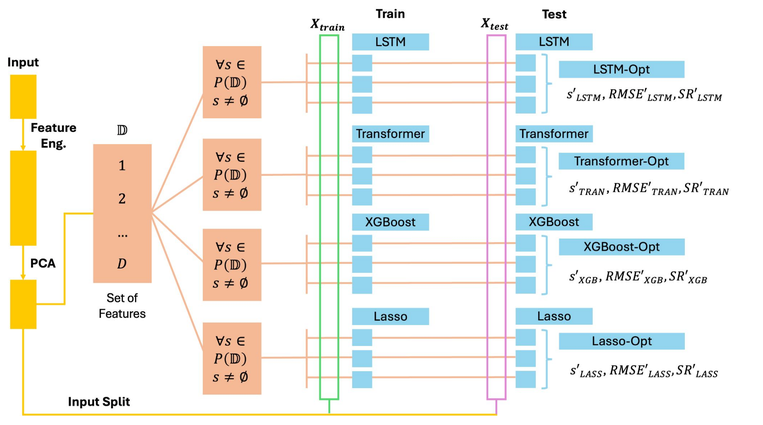
Methods

LSTM: Recurrent neural network that captures time-dependent patterns in stock price data, which is essential for discerning trends over time; LSTM memory cells learn both short-term details and long-term dependencies.

XGBoost: Robust ensemble of cross-validated decision trees that enhances prediction accuracy through boosting techniques; XGBoost optimizes computational speed and model performance by handling missing data and applying regularization. Cross-validation helps tune hyperparameters: number of trees, max. depth of a tree, minimum loss reduction for partition, learning rate.

Transformer: RNN with the self-attention mechanism to assess the importance of different parts of the input, which is well-suited for modeling stock prices. The model enhances training speed and scalability by processing data in parallel.

LASSO: Applies a L1 penalty with cross validated parameter to the regression coefficients, effectively reducing the number of variables the model relies on. We used cross-validation to determine the best regularization parameter.



Feature Engineering

Feature Creation:
Basic stock features: open, close, high, low, percentage returns, volatility
Technical indicators: RSI, MACD, BIAS
Moving averages: regular (MA), exponential (EMA)
Other: amplitude, open-to-close, 52 week high, expected dividends

Principal Component Analysis:
Minka's MLE[1] to predict the number of dimensions for PCA

Key Feature Calculation

$$RSI_x = \frac{\text{Average Gain During Up Period } x}{\text{Average Loss During Down Period } x}$$

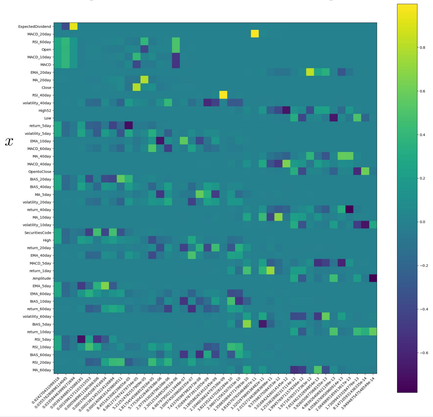
$$BIAS_x = \frac{\text{Closing Price} - x \text{ Day Average Price}}{x \text{ Day Average Price}}$$

$$MACD_x = \text{Exponentially Moving Average of Period } x - \text{Exponentially Moving Average of Period } 2x$$

$$Expected\ Dividend = I[\text{Expected Dividend} > 0]$$

Key technical indicators were determined from recent survey[2]. Expected dividend is recorded two days before ex-dividend date, which lines up with target date

PCA Component and Feature Importance



Results

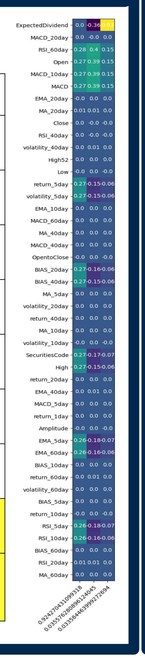
Principal Component Analysis: Three most significant components accounted for 99% of the variance

Important Features: RSI over 5, 10, and 20 days, BIAS over 20, and 40 days, MACD over 10 and 12 days, volatility over 5 days

Impact of Expected Dividends: We can see that the third component is explained mostly by the Expected Dividend feature. However, it does not explain most of the dataset, so the rest of the technical indicators are necessary

Sharpe Ratios Across Models and Feature Subsets

	LSTM	XGB	LASSO
PCA (0,1,2)	-0.036	0.132	0.090
PCA (0,1)	0.059	0.133	0.119
PCA (0,2)	-0.029	0.05	0.034
PCA (1,2)	0.04	0.085	0.106
PCA (0)	0.056	0.029	0.034
PCA (1)	0.126	0.064	0.106
PCA (2)	-0.086	0.009	0.003
Best w/o PCA	**0.116**	**0.132**	**0.053**
Best w/ PCA	**0.126**	**0.133**	**0.119**



Discussion

LSTM: Using PCA can lead to varied outcomes, but the best-performing subset exceeds the baseline (full features). The second principal component (PC) is the most predictive and leads to Sharpe ratio (SR) of **0.126** (Competition **Rank 266**).

XGBoost: Using subsets from PCA has little to no effect on the the SRs of our predictions. The best SR obtained with the XGBoost algorithm **without** PCA is **0.132** (Competition **Rank 251**), and the best SR obtained **with** PCA uses the the first two PCs, achieving a SR of **0.133 (**almost the same SR value as all three PCs)**.**

LASSO: The best-performing subset exceeds the baseline by a significant margin (more than double). The first two PCs are the most predictive and leads to SR of **0.119** (Competition **Rank 271**).

We see that using a subset of the three main PCs in the data **improves the performance of LSTM and LASSO models while having little impact on XGB.** The best subset was the same for both XGB and LASSO, and it was the subset with the first two principal components. The best subset for LSTM was subset containing only the second principal component. We believe that using subsets of features worked well because it reduces overfitting and allows the models to focus on the most relevant information.

Conclusion

Challenge: Throughout initial exploration, we found feature engineering and selection to have unreliable impacts on performance, with a degree of random variation that's difficult to account for.

Learnings: We can confirm Hypothesis (1), based on all three algorithms of LASSO, XGB and LSTM, meaning that is is valuable to try various subsets of features and seek the best set. We can also confirm Hypothesis (2), as the best subsets of features were not the same across all three models.

Limitations: There are numerous financial indicators that can be utilized given more time. Comparing subsets is a method that is limited by the number of features kept, since attempting too many subsets would be infeasible.

Future work: We plan to implement a Transformer model soon with the same feature engineering and subset selection method; we'll compare this with the LSTM model (another RNN model) to see how much it improved upon it

References

1. Minka, Thomas. Automatic Choice of Dimensionality for PCA. 2000.
2. H. H. Htun, M. Biehl, and N. Petkov. Survey of Feature Selection and Extraction Techniques for Stock Market Prediction. 2023.

Kaggle Competition:
https://www.kaggle.com/competitions/jpx-tokyo-stock-exchange-prediction

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